



Robotics – Modelling of Hexabot leg
Report

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Modelling of Mechanical Systems

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1 Introduction

The field of robotics has seen significant advancements in the last few decades, with hexapods or six-legged robots being of notable interest due to their stability and versatility in various terrains. A critical aspect of such robots is the kinematic understanding of individual legs. In this report, we delve into the modelling of a single hexapod leg with three revolute joints. By understanding the kinematics, we aim to determine the leg's elliptic movement to generate precise and consistent steps.

Our objective is to regulate three specific degrees of freedom for the foot. These degrees encompass translation along the x-axis, translation along the y-axis, and translation along the z-axis. we will study these frames of the hexabot leg.

The Denavit-Hartenberg (DH) parameterization is a systematic approach to describe the kinematics of a robotic arm or leg, simplifying the process of obtaining transformation matrices between links. The DH method introduces specific conventions to attach reference frames to the links of a robot.

1.1 DH Coordinate Frame Assumptions.

The beauty of the DH method is its consistent way of attaching coordinate frames to robot links based on these two assumptions:

1. The axis X_1 is perpendicular to the axis Z_0
2. The axis X_1 intersects the axis Z_0

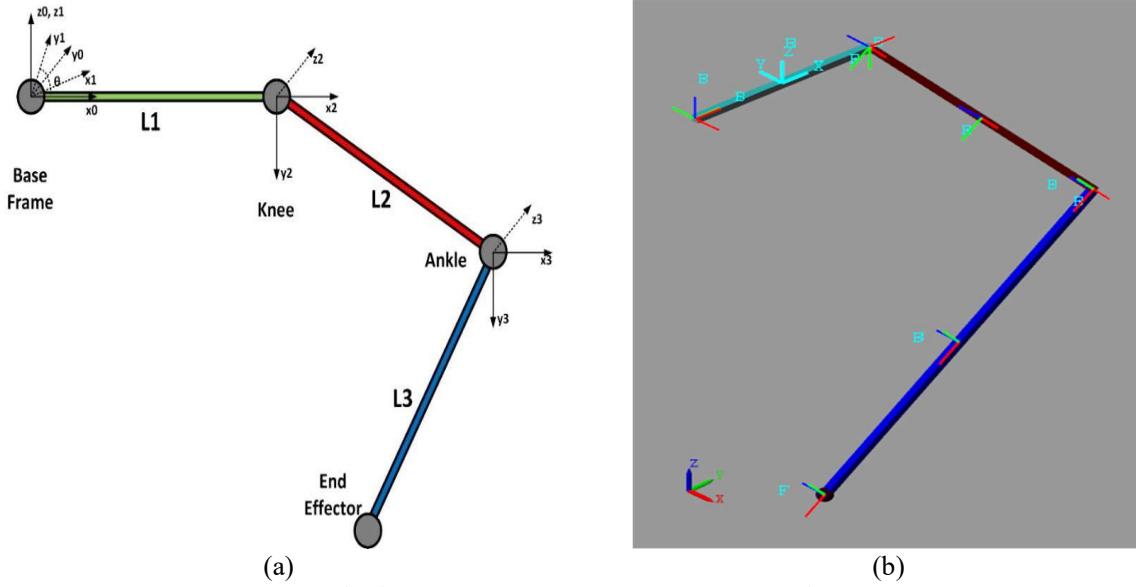


Fig 1: Hexabot leg model Frame representation

Using the DH parameters and Khalil formalism we obtain this Table 1:

Table 1: Frames obtained from DH / Khalil formalism

Frame Number	Type of Joint α	Distance along X_{i-1} d	Angle around X_{i-1} α	Distance along Z_{i-1} r	Angle around Z_{i-1} θ
1	0	0	0	0	θ_1
2	0	L_1	$\pi/2$	0	θ_2
3	0	L_2	0	0	θ_3
4	-	L_3	0	0	0

2 Transform Matrix

We will now calculate the homogenous transform matrix and the genral from matrix :

2.1 From Zero to 1st frame:

$$A = \begin{bmatrix} \cos(\theta_1) & -\sin(\theta_1) & 0 & 0 \\ \sin(\theta_1) & \cos(\theta_1) & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

2.2 From 1st frame to 2nd frame:

$$B = \begin{bmatrix} \cos(\theta_2) & -\sin(\theta_2) & 0 & L_1 \\ 0 & 0 & -1 & 0 \\ \sin(\theta_2) & \cos(\theta_2) & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

2.3 From 2nd to 3rd frame:

$$C = \begin{bmatrix} \cos(\theta_3) & -\sin(\theta_3) & 0 & L_2 \\ \sin(\theta_3) & \cos(\theta_3) & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

2.4 From 3rd 4th frame:

$$D = \begin{bmatrix} 1 & 0 & 0 & L_3 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

2.5 From Zero to 4th frame:

$$R = A \times B \times C \times D$$

$$= \begin{bmatrix} \cos(\theta_1) \cdot \cos(\theta_2 + \theta_3) & -\cos(\theta_1) \cdot \sin(\theta_2 + \theta_3) & \sin(\theta_1) & L_1 \cdot \cos(\theta_1) + L_2 \cdot \cos(\theta_1) \cdot \cos(\theta_2) + L_3 \cdot \cos(\theta_1) \cdot \cos(\theta_2 + \theta_3) \\ \sin(\theta_1) \cdot \cos(\theta_2 + \theta_3) & -\sin(\theta_1) \cdot \sin(\theta_2 + \theta_3) & -\cos(\theta_1) & L_1 \cdot \sin(\theta_1) + L_2 \cdot \sin(\theta_1) \cdot \cos(\theta_2) + L_3 \cdot \sin(\theta_1) \cdot \cos(\theta_2 + \theta_3) \\ \sin(\theta_2 + \theta_3) & \cos(\theta_2 + \theta_3) & 0 & L_2 \cdot \sin(\theta_2) + L_3 \cdot \sin(\theta_2 + \theta_3) \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Last column of homogenous matrix is position vector we can calculate the Jacobian from direct method :

Following are the position vectors :

$$P_x = L_1 \cdot \cos(\theta_1) + L_2 \cdot \cos(\theta_1) \cdot \cos(\theta_2) + L_3 \cdot \cos(\theta_1) \cdot \cos(\theta_2 + \theta_3)$$

$$P_y = L_1 \cdot \sin(\theta_1) + L_2 \cdot \sin(\theta_1) \cdot \cos(\theta_2) + L_3 \cdot \sin(\theta_1) \cdot \cos(\theta_2 + \theta_3)$$

$$P_z = L_2 \cdot \sin(\theta_2) + L_3 \cdot \sin(\theta_2 + \theta_3)$$

So, we calculate the derivative of P_x , P_y and P_z w.r.t θ_1 , θ_2 , and θ_3

2.6 Jacobian Matrix

Jacobain is:

$$J = \begin{bmatrix} -L_1 \cdot \sin(\theta_1) - L_2 \cdot \sin(\theta_1) \cdot \cos(\theta_2) - L_3 \cdot \sin(\theta_1) \cdot \cos(\theta_2 + \theta_3) & L_2 \cdot \cos(\theta_1) \cdot \sin(\theta_2) - L_3 \cdot \cos(\theta_1) \cdot \sin(\theta_2 + \theta_3) & -L_3 \cdot \cos(\theta_1) \cdot \sin(\theta_2 + \theta_3) \\ L_1 \cdot \cos(\theta_1) + L_2 \cdot \cos(\theta_1) \cdot \cos(\theta_2) + L_3 \cdot \cos(\theta_1) \cdot \cos(\theta_2 + \theta_3) & -L_2 \cdot \sin(\theta_1) \cdot \sin(\theta_2) - L_3 \cdot \sin(\theta_1) \cdot \sin(\theta_2 + \theta_3) & -L_3 \cdot \sin(\theta_1) \cdot \sin(\theta_2 + \theta_3) \\ 0 & L_2 \cos(\theta_2) + L_3 \cos(\theta_2 + \theta_3) & L_3 \cos(\theta_2 + \theta_3) \end{bmatrix}$$

So we have 3×3 jacobain , to get the simulation we need to put this jacobian in the MATLAB Function inside SIMULINK model.

3 Simulation Model

For simplicity we split the model into sub-blocks. The screenshots of the models are following:

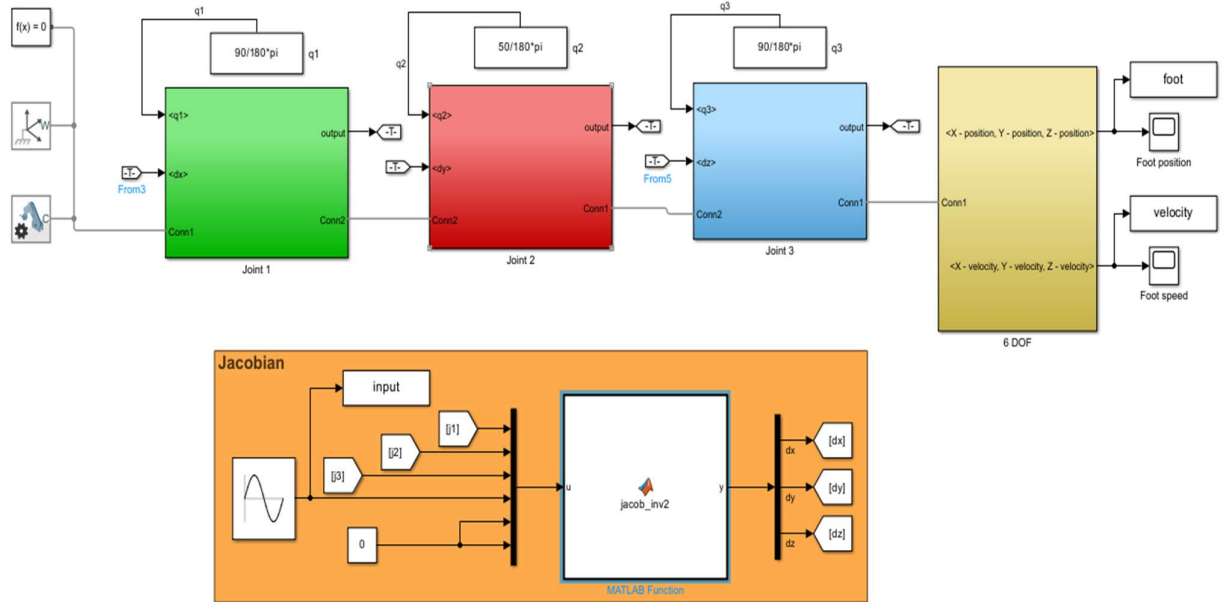


Fig 2: Simulink Model of Hexabot leg

Following Fig 3. shows the internal structures of joint 1, joint 2 and joint 3 while Fig 4. is for 6 DOF structure.

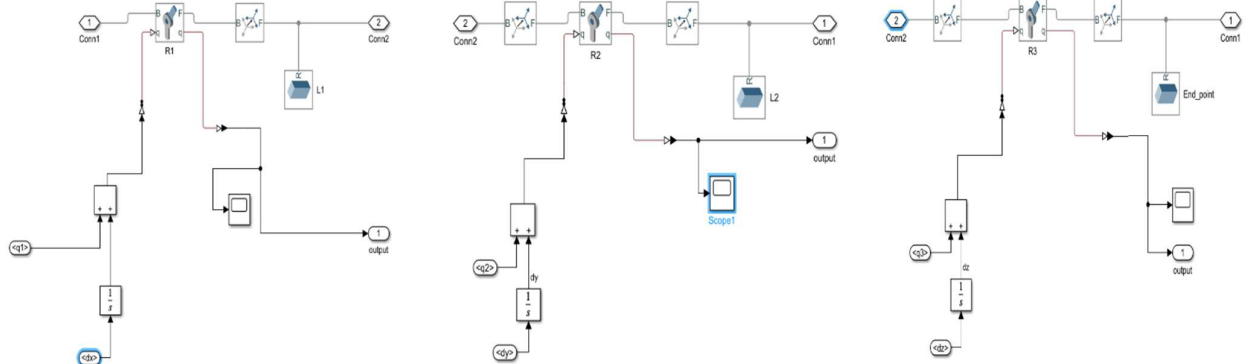


Fig 3: Screenshots of Joint 1, Joint 2 and Joint 3

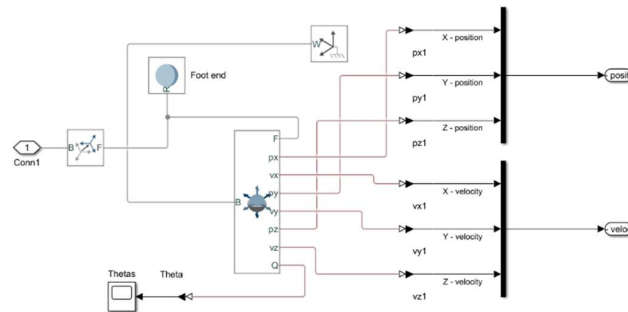


Fig 4: 6 Degrees of Freedom block of SIMULINK model

4 Results

Upon analysing the foot's movement, it becomes evident that its positional shifts occur exclusively along the X-axis. In contrast, both the Y and Z axes remain unaffected, showcasing a consistent positional steadiness with no observable velocity alterations in those directions. This behaviour underscores the system's ability to achieve a purely forward stepping motion, eliminating any extraneous or lateral body movements. The unequivocal evidence for this unique motion characteristic is provided by the aforementioned position and velocity simulation results

The following fig 5(a). shows the foot trajectory, while fig 5(b) is the detailed graph of foot trajectory:

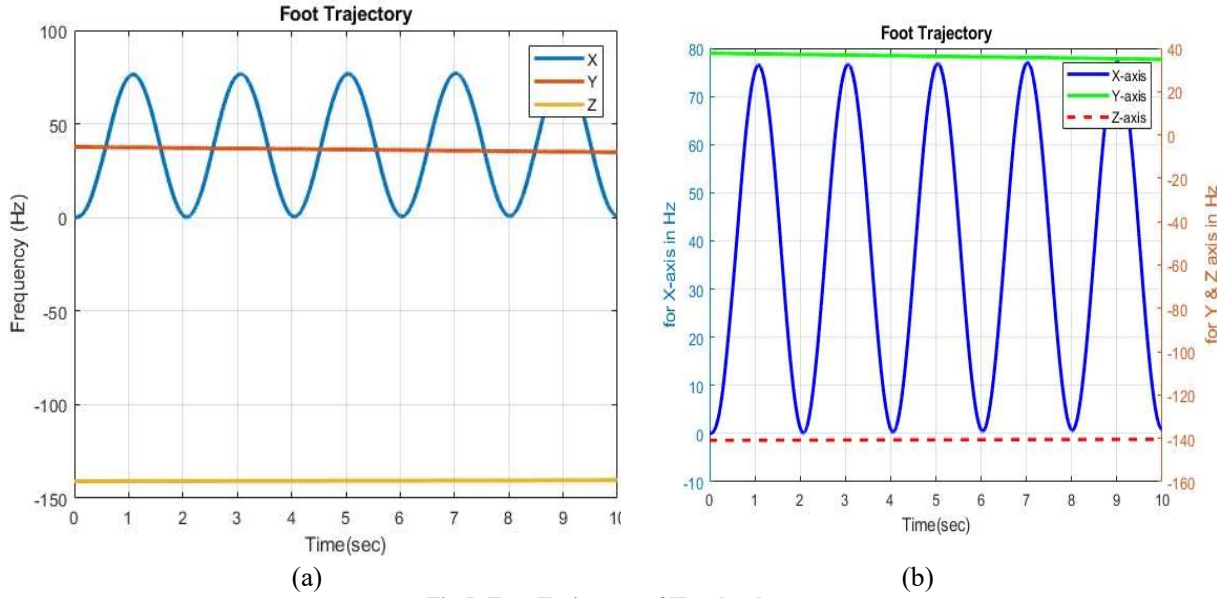


Fig 5: Foot Trajectory of Hexabot leg

The following fig 6(a). shows the velocity along x-axis, while fig 6(b) is the detailed graph of velocity:

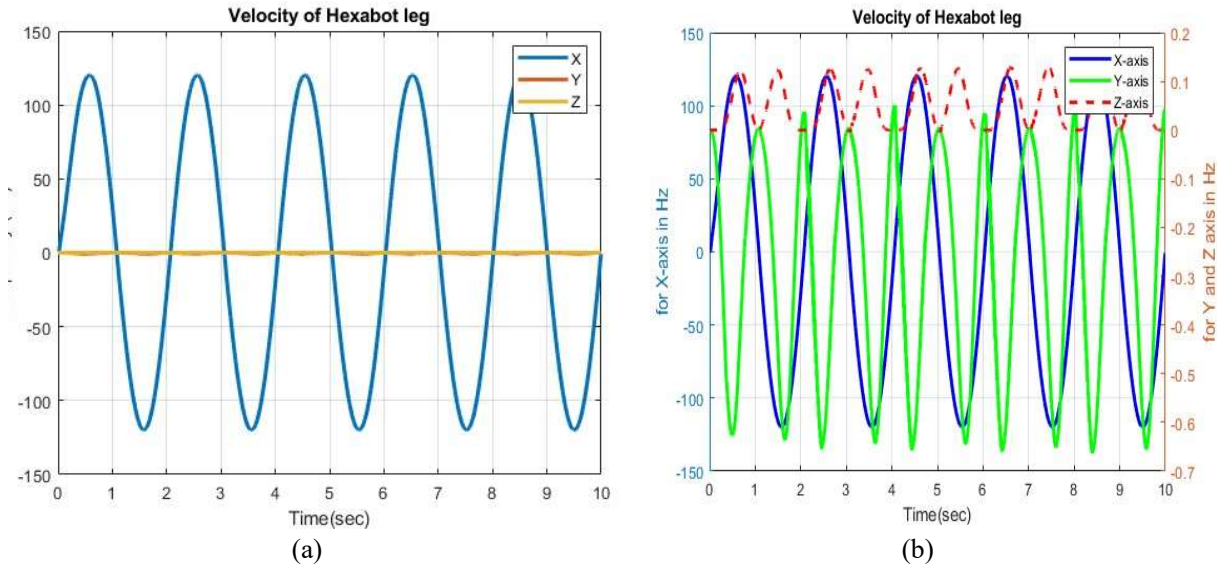


Fig 6: Velocity of Hexabot leg

The input sinusoidal signal with 120 Hz frequency is shown in following fig 7.

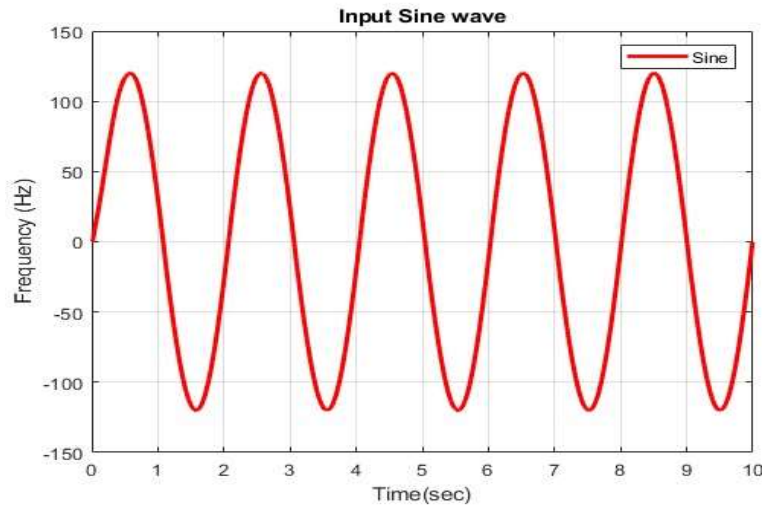


Fig 7: Input sinusoidal signal for motion.

The following fig 8. Shows the graph of angles of 6 DOF block, while fig 9 shows the motion:

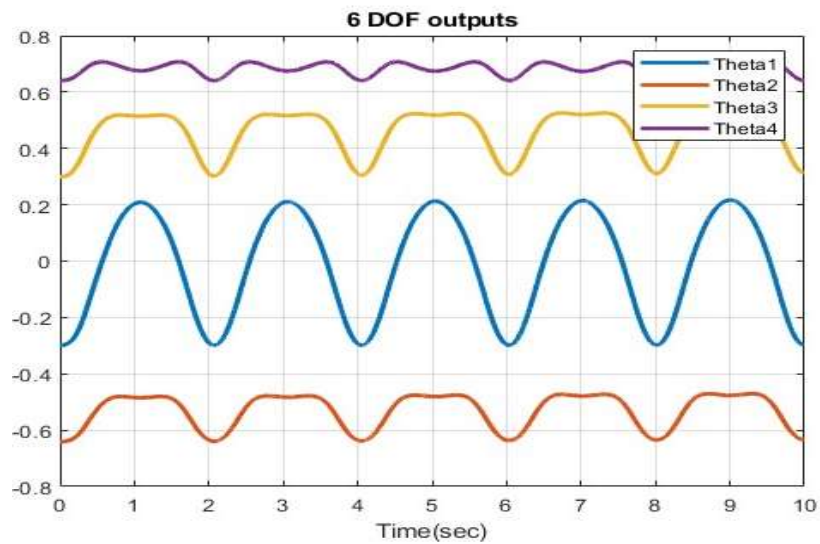


Fig 8: Output graph of 6 DOF block

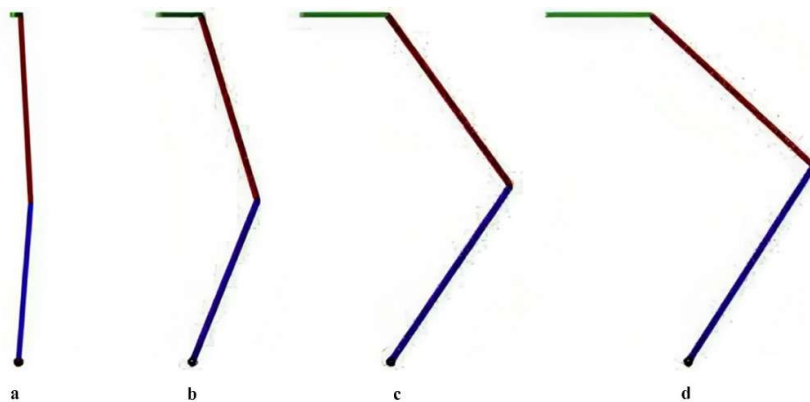


Fig 9: Motion of Hexabot leg